

PAPER

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2024, 17, 3375"Freezing" intermediate phases for efficient and stable FAPbI₃ perovskite solar cells†Muyang Chen,^{‡a} Tingting Niu,^{‡a} Lingfeng Chao,^{id a} Xiaozheng Duan,^{id b} Jingpei Wang,^a Tengfei Pan,^a Yajing Li,^a Junhan Zhang,^c Chenyue Wang,^c Biyun Ren,^a Lijuan Guo,^a Mohammad Hatamvand,^{id a} Jing Zhang,^d Qingxun Guo,^a Yingdong Xia,^{id a} Xingyu Gao^c and Yonghua Chen^{id *a}

The most promising formamidinium lead triiodide (FAPbI₃) has attracted extensive attention due to its record-breaking single-junction cell efficiency in perovskite solar cells (PSCs). However, the formation of α -FAPbI₃ often requires δ -FAPbI₃ as an intermediate phase, which suffers from low crystallinity and disordered crystal orientation, resulting in a high defect density in the final α -FAPbI₃ phase. Herein, we report the synthesis of a stabilized and high crystallinity intermediate phase through full process (solution, anti-solvent, substrate, and processing atmosphere) low-temperature control. The low-temperature strategy allows the formation of larger and more stable colloids in the precursor solution for fewer but superior initial nuclei, and retards the film crystallization for better intermediate phase crystals. The defect density of the final perovskite thin film was reduced by 8 times. As a result, the efficient and stable FAPbI₃ PSC was obtained with an efficiency of over 25% and a fill factor of over 85%, along with excellent long-term operating stability for 1000 hours under continuous light stress.

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Broader context

Formamidinium lead iodide perovskite (FAPbI₃) is considered an ideal metal halide perovskite semiconductor for high-efficiency single-junction perovskite solar cells (PSCs) due to its narrow bandgap (1.47 eV). However, the high density of bulk defects in the FAPbI₃ perovskite film significantly limits its further efficiency improvement. One of the main reasons for the high density of bulk defects in FAPbI₃ thin films is the strong stress changes caused by the δ to α phase transition process, since the intermediate phase often has chaotic crystal orientation and low crystallinity. This work is based on a direct tailoring processing temperature method. The bulk defect density of the α -FAPbI₃ thin film prepared by this method has decreased by about 8 times compared to traditional methods and thus got a high efficiency of 25.05%. The core of this method is to reduce the process temperature to accurately control crystal nucleation and growth. This strategy can contribute significantly to PSCs by reducing the problems caused by the density of bulk defects in perovskite films.

Introduction

Due to their ideal bandgap, FAPbI₃ perovskite solar cells (PSCs) have the highest theoretical efficiency in single-junction PSCs

and are considered a strong challenger to current state-of-the-art photovoltaic techniques.^{1,2} At present, the certified efficiency of FAPbI₃ PSCs has increased to 26.1%, which is close to the record of single-crystal silicon solar cells.^{3–7} As a polycrystalline thin-film structure, high-quality FAPbI₃ perovskite thin films with low defect density are necessary to drive efficiency to reach the Shockley–Queisser limitation.^{8,9} Therefore, many studies of the preparation of FAPbI₃ thin films have been carried out to improve the crystallization and phase formation.^{5,10–12} Ion doping^{13–15} and additive engineering^{16–21} are both widely used methods to regulate crystallization and assist in the formation of α -FAPbI₃, in which MA⁺, Cs⁺, and Br[−] ions are often used to control the perovskite crystallization kinetics, while MAcl, MDacl₂, and other additives regulate the intermediate phase. Moreover, solvent engineering and vapor induction engineering have also been explored.^{22–26} However,

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these studies pay much attention to the phase transition processes while neglecting the intermediate δ -FAPbI₃ phase formation process, which is also important for the final α -FAPbI₃ crystal quality and especially the defect density. To improve the crystal quality of the intermediate δ -FAPbI₃ phase, temperature management might actively influence the morphology of the perovskite single crystal.^{27,28} It is worth noting that lower temperature is believed to help reduce defects in perovskite crystals.²⁹ However, very little is known about how tailoring the temperature both in solution and film formation might translate into the overall enhancement of the intermediate phase crystal and the following improvement of efficiency and stability of the perovskite thin films.

Here, we report the synthesis of FAPbI₃ perovskite thin films with super low defect density through a full process (solution, anti-solvent, substrate, and processing atmosphere) low-temperature control (Fig. S1, ESI†). We found that the precursor solution of FAPbI₃ has fewer and higher quality nucleation sites at a low-temperature of -20°C . The low-temperature strategy further allows retarding the crystallization process, thus leading to the production of a stable intermediate δ -FAPbI₃ phase with higher crystallinity and initial α -FAPbI₃ nuclei inside. After annealing, the final α -FAPbI₃ shows an extremely low defect density reduced by 8 times. Using this low-temperature deposition method, we obtained an efficient and stable FAPbI₃ PSC with an efficiency of over 25% and a fill factor of over 85%,

along with excellent long-term operating stability for 1000 h under continuous light stress.

Results and discussion

Precursor solution chemistry at low temperature

The precursor solution chemistry plays an important role in perovskite nucleation and crystallization at different temperatures. An obvious Tyndall effect was observed in the precursor solutions at -20°C , while no such effect was found at 20°C and 60°C (Fig. S2A–C, ESI†). We note that the solidification of DMSO in a mixed solvent (DMF:DMSO = 8:1) at low temperatures could be ruled out because no Tyndall effect in the pure mixed solvent could be observed (Fig. S3A and B, ESI†). The simulation results further confirmed that no DMSO clusters formed through aggregation at -20°C (Fig. S3C–E, ESI†), Demonstrating that the perovskite colloidal size was significantly increased with decreasing the temperature (Fig. S4, ESI†). The precursor has a massive colloidal size of roughly 1740 nm at -20°C compared to 562 nm at 20°C and 152 nm at 60°C . The large colloidal size has been suggested to favor favorable spontaneous crystallization in the film formation.³⁰ To investigate the evolution of chemical interaction in precursor solutions at different temperatures, the extended X-ray absorption fine structure (EXAFS) was used to observe the coordination around the lead ion²² (Fig. 1A–C and Fig. S5, ESI†). The first

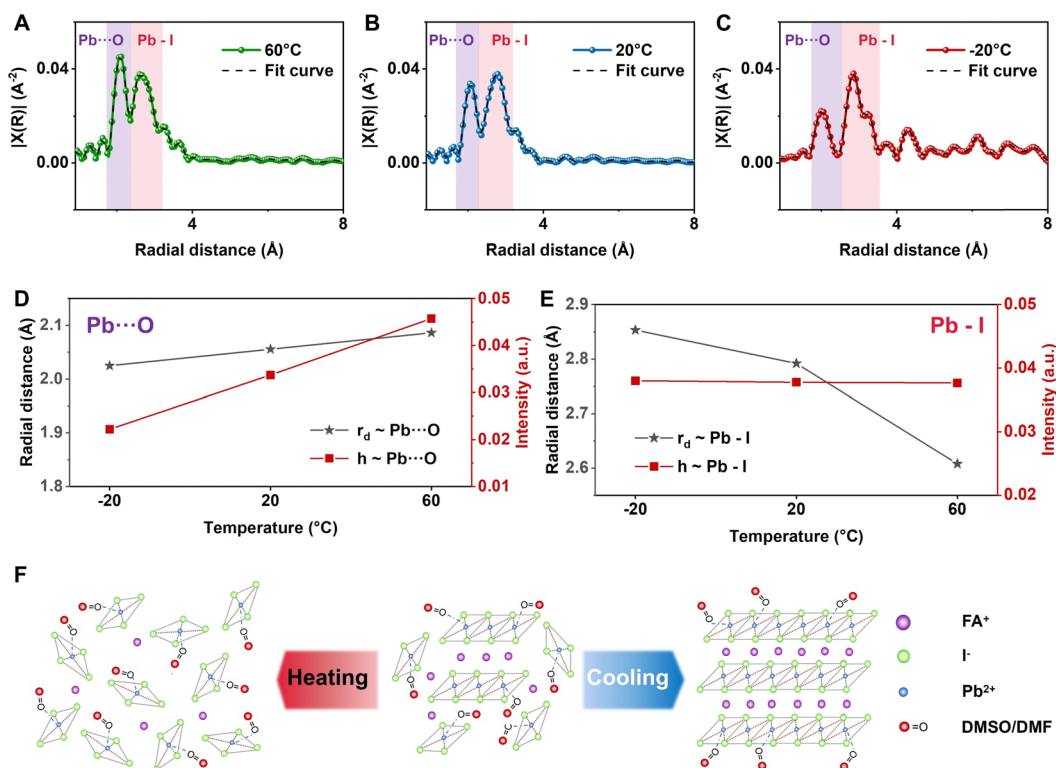


Fig. 1 Effect of temperature on the colloidal state of the FAPbI₃ precursor. Fourier-transformed radial distribution function of the Pb L3-edge extended XAFS spectra in R-space for the FAPbI₃ precursor colloidal solution at temperatures of (A) $T \sim 60^\circ\text{C}$, (B) $T \sim 20^\circ\text{C}$ and (C) $T \sim -20^\circ\text{C}$ (concentration is diluted to 0.3 M). The variation of peak intensity and position of (D) the Pb···O peak and (E) the Pb-I peak with the decrease of the temperature. (F) Schematic diagram of the FAPbI₃ precursor colloid and the influence of the colloidal solution temperature on the Pb···O and Pb-I chemical bonds.

peak corresponding to the $\text{Pb}\cdots\text{O}=\text{C}$ or $\text{O}=\text{S}$ chemical bond shows a slight blue-shift from 2.08 Å to 2.05 Å and 2.02 Å for 60 °C, 20 °C, and −20 °C (Fig. 1D), respectively, indicating the stronger interaction between the Pb–I framework and DMF:DMSO solvent at lower temperature. Besides, the intensity of the $\text{Pb}\cdots\text{O}=\text{C}$ or $\text{O}=\text{S}$ peak significantly decreased, indicating a decrease in the number of solvent molecules (DMF, DMSO) coordinating with Pb, because DMF and DMSO mainly coordinate with Pb on the colloid surface, and the size growth of the colloid surface is lower than the volume growth of the colloid. This corresponded to an increase in colloid size due to the corner-shared octahedral soft Pb–I framework growth. The second peak corresponding to the Pb–I chemical bond appears at 2.86 Å for −20 °C, 2.8 Å for 20 °C, and 2.6 Å for 60 °C (Fig. 1E),^{31,32} respectively, indicating that the Pb–I bonds in the PbI_6^{4-} framework are stretched in the precursor. This is because FA^+ is easy to get into the negatively charged PbI_6^{4-} frameworks at low temperatures to balance positive and negative charges, thus expanding the soft framework. As a result, the PbI_6^{4-} and FA^+ form larger colloids for fewer and higher quality initial crystal nuclei of the perovskite at lower temperatures, which could contribute to forming high-quality FAPbI_3 perovskite thin films. On the contrary, the PbI_6^{4-} soft framework becomes more fragmented at higher temperatures, which is not conducive to the entry of cations into the framework. A schematic diagram shows the change in the chemical environment around the Pb–I frameworks at different temperatures (Fig. 1F).

Molecular dynamics simulation

Molecular dynamics (MD) simulation can be used as an effective approach to analyze the detailed physicochemical features of the ionic solutions on the molecular level. To understand the underlying mechanisms for ionic clusters in the precursor solution, we further employ coarse-grained MD simulations to study the microscopic properties of the $\text{PbI}_2/\text{FAI}/\text{MAI}/\text{DMSO}/\text{DMF}$ mixed solution used for perovskite crystallization. Herein, we develop the coarse-grained model based on our previous simulations^{33–39} and other ref. 40. The simulations are performed *via* the large-scale atomic/molecular massively parallel simulator (LAMMPS),⁴¹ and the details of the model (Fig. S6, ESI†) and simulation method are provided in the ESI.†

We display the typical simulation snapshots of the precursor solution in Fig. 2A–C and show the corresponding structural animation in Movie S1 (ESI†), where temperature is decreased from $T \sim 60$ °C to $T \sim -20$ °C. Furthermore, we calculate static structural factors of ionic species $S_i(q)$ (Fig. 2D), the radial distribution functions of I^- around each Pb^{2+} $g_{\text{Pb-I}}(r)$ (Fig. 2E), monovalent cations (FA^+ and MA^+) around each I^- $g_{\text{I-FA/MA}}(r)$ (Fig. S6A, ESI†) or around each Pb^{2+} $g_{\text{Pb-FA/MA}}(r)$ (Fig. S7B, ESI†), solvent molecules (DMSO and DMF) around each I^- $g_{\text{I-S}}(r)$ (Fig. S6C, ESI†), the number of neighboring monovalent cations $n_{\text{I-FA/MA}}$ and solvent molecules $n_{\text{I-S}}$ around each I^- (Fig. 2F) and the diffusion coefficients of each component D (Fig. 2G), to characterize the structural and dynamic variations of the solution induced by temperature decrease. Our simulations indicate that at a higher

temperature $T \sim 60$ °C, the ionic species weakly aggregate, as shown by the snapshots (Fig. 2A), Video S1 (ESI†) and peaks in $S_i(q)$ (Fig. 2D). In this context, the Pb^{2+} ions strongly bind with the I^- ions, whereas the monovalent cationic FA^+ and MA^+ show relatively weak electrostatic interactions with I^- , as indicated by $g_{\text{Pb-I}}(r)$ (Fig. 2E) and $g_{\text{I-FA/MA}}(r)$ (Fig. S7A, ESI†). Due to the excessive amount of I^- , Pb^{2+} and I^- only form small clusters dispersed in the solutions. With decreasing the temperature from $T \sim 60$ °C to $T \sim -20$ °C, the ions and solvent molecules exhibit much slower diffusion, as shown by the decreased D (Fig. 2G). In this process, the potential energy overwhelms the kinetic energy, and therefore, the ions tend to combine with each other through relatively stronger electrostatic interactions, which significantly enhances ionic clusters (Fig. 2D and E). In particular, our simulation demonstrates that at low temperatures, the monovalent cationic FA^+ and MA^+ ions can further bridge between the Pb/I clusters, and this effect enables the formation of large ionic clusters (Fig. 2D, F and Fig. S7A, B, ESI†). Accordingly, the ionic species show less affinity with the solvent molecules due to the clustering (Fig. 2F and Fig. S7C, ESI†). These results are consistent with the enhanced Pb–O (*i.e.*, Pb– FA^+/MA^+) and increased ionic cluster affinity induced by programmed freezing as shown in Fig. 1D and E.

We thus conclude that freezing can cause significant slowing dynamics of each component and apparent aggregation of ionic species in the precursor solution and optimize the following perovskite crystallization. Specifically, at higher temperature (~ 60 °C), Pb^{2+} and I^- assemble into smaller clusters in solution and during the spin-coating process, the dispersed $\text{Pb}^{2+}/\text{I}^-$ clusters only form small crystalline grains after the solvent evaporation. However, at the freezing temperature of ~ -20 °C, Pb^{2+} and I^- aggregate into larger clusters bridged by the monovalent FA^+ and MA^+ ions, which can further form large-scale, connected crystals after the spin-coatings. Our simulations clarify the microscopic mechanisms for the distinct types of crystallization caused by the temperature variations of the precursor solutions.

Crystallization of perovskite thin films at various temperatures

To find out the influence of different temperatures on the crystallization of FAPbI_3 thin films, we use grazing incidence wide-angle X-ray scattering (GIWAXS) measurement to monitor the complete process from spin coating to annealing (Fig. 3A). The whole process is divided into four stages, which are marked by different colors. In stage I (0 s to 8 s, a spin starts to anti-solvent (ether) dripping), the solvent was gradually evaporated and the solution tended to supersaturation. In stage II (8 s to 22 s, anti-solvent dripping to spin ends), the 2H (δ - FAPbI_3) phase at 8.3 nm^{-1} is dominated in the films at all temperatures, and weak 4H (complex hexagonal phase) at 8.2 nm^{-1} , 6H (complex hexagonal phase) at 8.6 nm^{-1} and 3C (α - FAPbI_3) at 9.8 nm^{-1} signals also appeared at the same time under -20 °C and 20 °C (Fig. S8, ESI†), indicating that the transformation to α - FAPbI_3 has occurred even though no

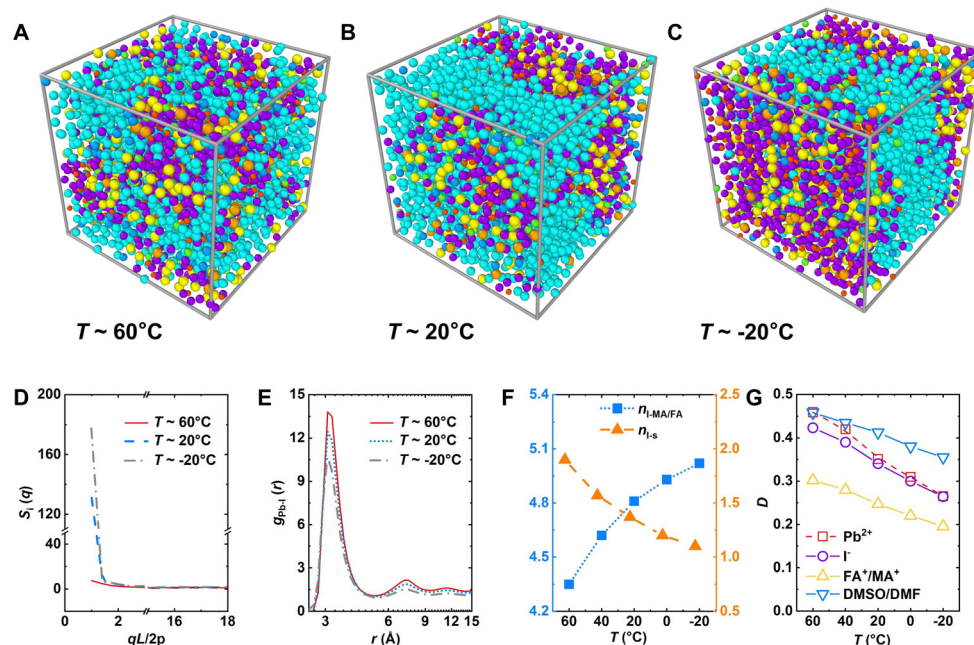


Fig. 2 Molecular dynamics simulation characterization. Typical simulation snapshots for the $\text{PbI}_2/\text{FAI}/\text{MACl}/\text{DMSO}/\text{DMF}$ precursor solution at temperatures of (A) $T \sim 60^\circ\text{C}$, (B) $T \sim 20^\circ\text{C}$ and (C) $T \sim -20^\circ\text{C}$, in which the red, purple, yellow, orange, green, dark blue and light blue beads indicate the Pb^{2+} , I^- , FA^+ , MA^+ , Cl^- , DMSO and DMF, respectively. (D) Static structural factor for ionic species at different temperatures. (E) Radial distribution functions of (E) I^- around each Pb^{2+} $g_{\text{Pb-I}}(r)$ at different temperatures. (F) Number of neighboring monovalent cations (FA^+ and MA^+) $n_{\text{I-FA/MA}}$ and solvent molecules (DMSO and DMF) $n_{\text{I-S}}$ surrounding each I^- at different temperatures. (G) Diffusion coefficients of each component at different temperatures.

annealing was performed. At 60°C , however, the precursor solution is unable to generate clusters of sufficient size to allow FA^+ to enter the PbI_6^{4-} framework. This results in the initial $\delta\text{-FAPbI}_3$ grain size being too small to produce a small amount of $\alpha\text{-FAPbI}_3$ inside it. This was further observed in the films placed in ambient air at room temperature for 30 minutes (Fig. S9, ESI†). The formation of $\alpha\text{-FAPbI}_3$ seeds without annealing has been suggested to be beneficial to the further growth of crystals.^{42,43} As a comparison, for the FAPbI_3 perovskite thin film without MACl, the $\delta\text{-FAPbI}_3$ phase in the final FAPbI_3 perovskite thin film was also significantly suppressed, while the intensity of the $\alpha\text{-FAPbI}_3$ phase increased (Fig. S10, ESI†). This indicates that the $\alpha\text{-FAPbI}_3$ seeds produced at low temperatures, instead of MACl, played a significant role in the phase transition process. Furthermore, a diffraction ring appeared at $q \approx 8.3 \text{ nm}^{-1}$ at all temperatures in the 2D GIWAXS images when $t = 20 \text{ s}$ were intercepted but the signal intensity increased with the decrease of the temperature (Fig. 3B–D), indicating that the $\delta\text{-FAPbI}_3$ phase has higher crystallinity at low temperature. This was further confirmed by the one-dimensional azimuth integral of the two-dimensional *in situ* GIWAXS diffraction pattern of the (100) plane of the $\delta\text{-FAPbI}_3$ with stronger orientation at 90° at -20°C (Fig. S11, ESI†).

In stage III (22 s to 42 s, heat to 150°C), the film was heated to 150°C from its original temperature. The 2H peak of the films formed at 20°C and 60°C disappeared immediately and quickly transformed into weak peaks belonging to 4H and 6H. In the end, these peaks disappeared and a 3C peak occurred. In addition, it can be noted that the 3C phase that occurred in

stage II at 20°C disappeared immediately because the crystal size and quality of the intermediate phase at 20°C is still not sufficient to be retained during the process of heating and annealing. However, when the spin coating was completed at a low temperature of -20°C , the substrate and the film remained at a low temperature, which slowed down the phase transformation process and the high-quality $\delta\text{-FAPbI}_3$ phase can effectively protect the initial $\alpha\text{-FAPbI}_3$ from being damaged by the violent action of solvents during the heating process. We found that 2H, 4H, 6H, and 3C phases coexist in the entire heating process (Fig. S8, ESI†), which mitigates the dramatic changes in lattice stress, thereby reducing the possible crystal defects in this process.^{44,45} Accordingly, in stage IV (42 s to 240 s, 150°C annealing), the final FAPbI_3 thin films formed at a low temperature of -20°C show a better crystallinity, with a higher peak intensity in the 90° direction in the (100) plane (Fig. S12A–C and S13, ESI†). The same results were also obtained for a longer exposure time ($t = 40 \text{ s}$) (Fig. S14A–C and S15, ESI†).

It should be noted that the lattice strain was normally generated in the final FAPbI_3 crystal due to the phase transition from δ -phase to α -phase. We therefore set the α_i from 0.1° to 0.5° to check the lattice change from top to bottom⁴⁶ (Fig. 3E–G and Table. S1, ESI†). Three thin films coated at all temperatures show the same (100) peak position around 9.73 nm^{-1} when $\alpha_i = 0.5^\circ$. However, the (100) peak position shifts to small angles when the α_i decreases to 0.2° and 0.1° at 60°C and 20°C (Fig. 3E and F) while that shifts to big angles at -20°C . This difference shows that the lattice was contracted in the films at 60°C and 20°C while the lattice stress relaxed in the films at -20°C .

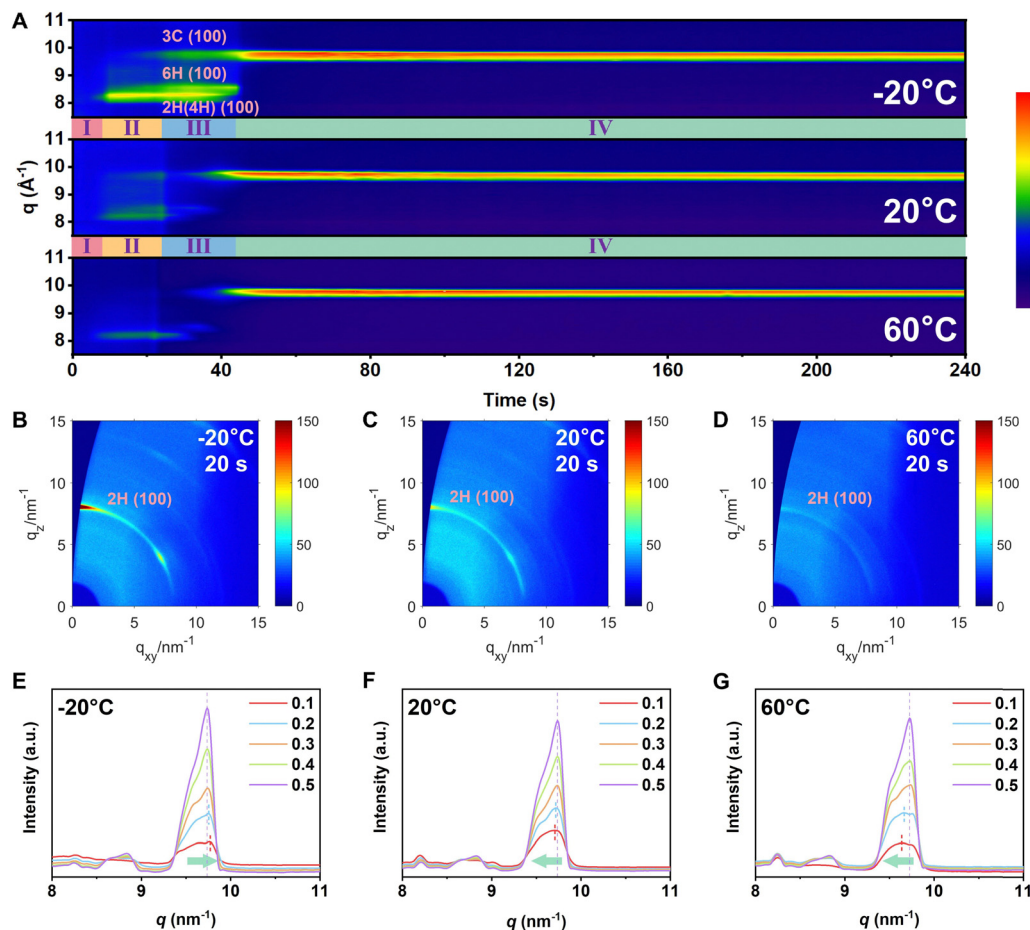


Fig. 3 Characterization of the FAPbI₃ crystallization and growth process. (A) *In situ* GIWAXS characterization of δ -phase perovskite formation at $T \sim -20$ °C, $T \sim 20$ °C and $T \sim 60$ °C, respectively, and the transformation from δ -phase to α -phase at 150 °C. Two-dimensional GIWAXS patterns of the perovskite film after dropping diethyl ether anti-solvent at temperatures of (B) $T \sim -20$ °C, (C) $T \sim 20$ °C and (D) $T \sim 60$ °C ($t = 20$ s, $\alpha_i = 0.5$, exposure for 2 s), intercepted from stage II of *in situ* GIWAXS characterization. Variable angle GIWAXS patterns from 0.1 to 0.5 degrees for FAPbI₃ thin films formed at temperatures of (E) $T \sim -20$ °C, (F) $T \sim 20$ °C, and (G) $T \sim 60$ °C (exposure for 40 s).

This can be attributed to the fact that many initial nuclei are formed at the top of the solution, which makes it easier to form small δ -FAPbI₃ crystalline grains on the film surface at higher temperatures (Fig. S16, ESI†). Finally, these small grains grow vigorously during the annealing process and extrude in the direction of the dominant growth orientation, while voids form in the direction of the non-dominant growth orientation. However, no excessive crystal nuclei on the surface were formed due to the low temperature in the films formed at -20 °C and the resulting vertical growth allows stress to be released from the surface of the film, leading to the relaxed lattice. On the contrary, if the growth angle of the perovskite crystals deviates from the vertical, the squeezing between adjacent grains will be more intense, resulting in greater surface lattice stress.

Performances of perovskite thin films and devices

The resulting FAPbI₃ thin films ($T = 20$ °C) exhibited better morphology and crystallinity than those fabricated at 20 °C

and 60 °C, with dense, large, and uniformly distributed crystals and smooth surfaces (Fig. S17 and S18, ESI†). This is beneficial for reducing the bulk defect density of the perovskite. The roughness of the perovskite thin films also decreased with the decrease in temperature (Fig. S19, ESI†). Cross-sectional scanning electron microscopy images further indicate the continuous large grains with few grain boundaries from the top to the bottom (Fig. S20, ESI†). The high crystallinity was further demonstrated by the reduced full width at half maxima of the (100) peak from 0.071 to 0.068 with decreasing the temperature (Fig. S21, ESI†). Moreover, the film formed at -20 °C over those at 20 °C and 60 °C exhibits a longer lifetime for non-radiative recombination (3218 ns) compared to those at 20 °C (1411 ns) and 60 °C (889 ns) (Fig. S22, ESI†). It can be considered that non-radiative recombination is significantly inhibited due to the improvement of film quality, as further confirmed by the improved photoluminescence intensity (Fig. S23, ESI†).

Having shown the significantly improved film quality with decreasing the temperature, we move on to investigate the performance of the corresponding PSCs. The device was fabricated with a structure of fluorine-doped tin oxide (FTO)/tin

oxide (SnO_2)/FAPbI₃/2,2',7,7'-tetrakis(*N,N*-dimethoxyphenylamine)-9,9'-spirobifluorene (Spiro-OMeTAD)/MoO₃/Au. The current density–voltage (J – V) characteristics of the devices based on films formed at different temperatures were performed. It should be noted that the device performance was so poor for 60 °C (Fig. S24, ESI†) that the devices with the films for –20 °C (Target) and 20 °C (Control) were intensively investigated. The best target device has a J_{sc} of 26.01 mA cm^{-2} (Fig. S25, ESI†), V_{oc} of 1.140 V, FF of 84.44% and PCE of 25.05% in reverse scan and J_{sc} of 25.99 mA cm^{-2} , V_{oc} of 1.138 V, FF of 83.86% and PCE of 24.81% in forward scan, while a J_{sc} of 25.88 mA cm^{-2} (Fig. S25, ESI†), V_{oc} of 1.114 V,

FF of 81.33 and PCE of 23.46% in reverse scan and J_{sc} of 25.90 mA cm^{-2} , V_{oc} of 1.101 V, FF of 79.94% and PCE of 22.81% in forward scan were achieved for the best control device (Fig. 4A and Table. S2, ESI†). As a comparison, the device ($T \sim -20$ °C) that employed only DMF showed an efficiency of 24.23% (Fig. S26, ESI†). The target devices show a high repeatability with an average efficiency of 24.16% (Fig. 4B, and Fig. S27A–C, ESI†). The statistical results indicate the major improvement parameters are V_{oc} and FF, which have a great relationship with defect recombination. It is worth noting that the highest FF of the PSC reaches 85.41% (Fig. S28, ESI†), indicating the improved balanced charge transport due to the

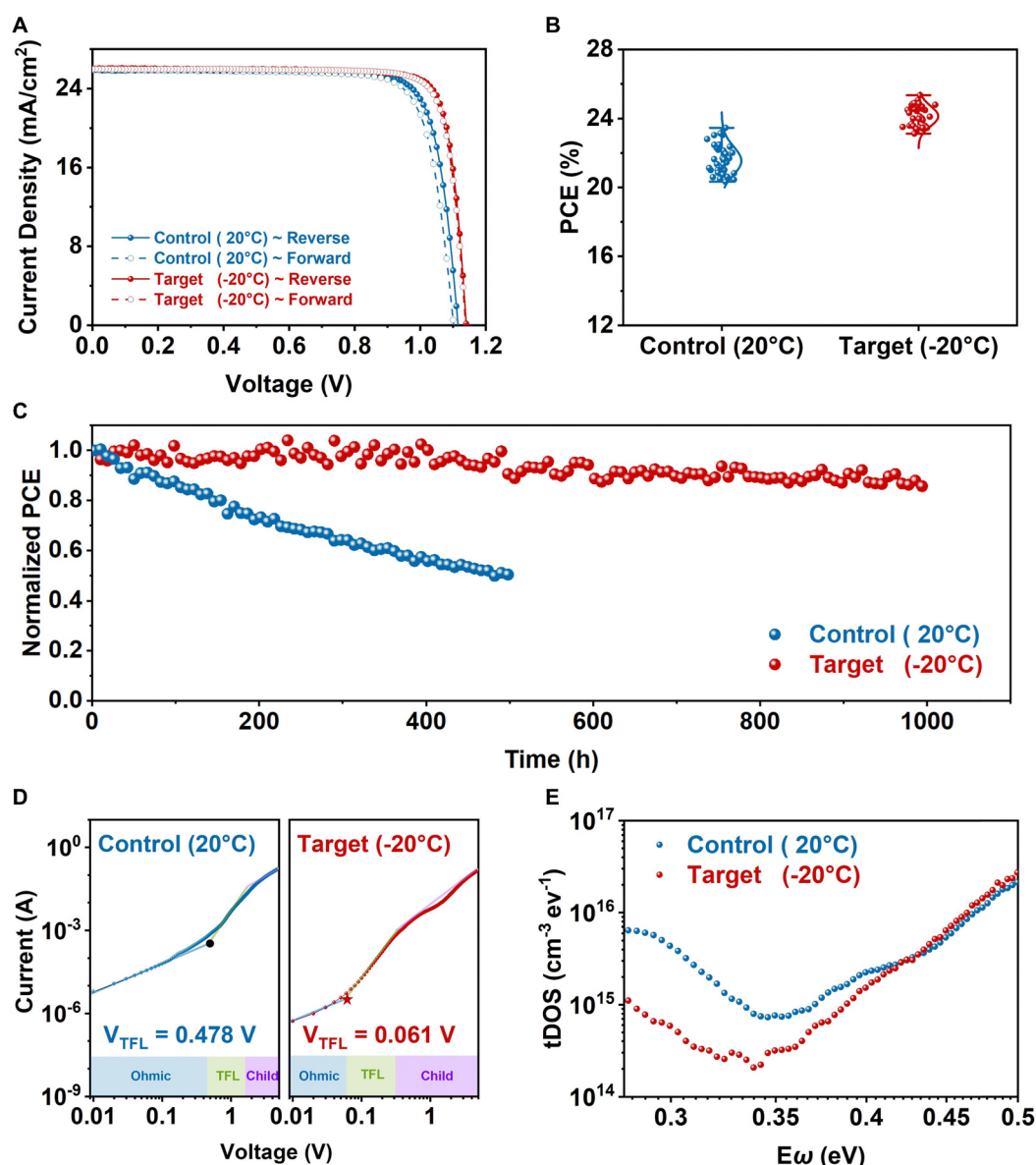


Fig. 4 Characterization of the PSC devices. (A) Forward and reverse scanning current–voltage (J – V) curves of the optimal control and target devices. (B) Statistical distribution of efficiency of 40 devices for control and target. (C) Long-term operational stability was measured by aging the encapsulated devices under an ambient atmosphere for 1000 h, using load resistance tracking under a white light-emitting diode lamp (AM 1.5G illumination). (D) The J – V curves of electron-only devices (FTO/ SnO_2 /FAPbI₃/PCBM/Ag) and extracted defect densities for the control and target FAPbI₃ perovskite thin film. (E) The energy-dependent trap density of states (TDOS) of control and target devices.

reduced defect density. The stable output for the target device (24.00 mA cm^{-2} , 1.01 V) over the control device (22.19 mA cm^{-2} , 0.97 V) indicates efficient charge extraction and negligible charge accumulation (Fig. S29, ESI†). Moreover, the PSCs exhibited long-term operational stability measured by aging the encapsulated devices under an ambient atmosphere, using load resistance tracking under a white light-emitting diode lamp (AM 1.5G illumination) (Fig. 4C). The PCE of the target device decreased by $\sim 10\%$ of its initial efficiency after 1000 h, while a 50% decrease was observed in the control device. Defect-induced degradation is often considered one of the reasons for the poor stability of PSCs.^{48,49} Thus, the improved stability is mainly due to the significantly improved film quality with reduced defect density.

The improved device performance was highly related to the reduced defect density in the perovskite films with low temperatures. The space charge limited current method was first used to confirm the reduction of the defect concentration in the crystal. The calculated target perovskite $N_{\text{trap}} = 8.79 \times 10^{14} \text{ cm}^{-3}$ is almost one order of magnitude less than the control perovskite $N_{\text{trap}} = 6.42 \times 10^{15} \text{ cm}^{-3}$ (Fig. 4D). It is worth noting that this is the lowest defect density in FAPbI₃ perovskite thin films as far as we know (Fig. S30, ESI†). Furthermore, the trap density of states (TDOS) of the target device significantly reduced over a trap depth below 0.4 eV, and was 8 times lower than that of the control device in the range from 0.3 eV to 0.35 eV (Fig. 4E). However, when the trap depth is higher than 0.40 eV, the change in TDOS between the control and target devices is not obvious. This is because the TDOS in FAPbI₃ perovskite assigned deeper traps ($> 0.40 \text{ eV}$) in the region to the defects at the film surface and shallower trap states ($< 0.40 \text{ eV}$) to the defects at grain boundaries.⁴⁷ This result indicates that low-temperature engineering can be particularly effective in reducing bulk defects in perovskite thin films. This was further confirmed by the lower leakage dark current density in the target device over the control device (Fig. S31, ESI†). The reduced defects were also observed by electrochemical impedance spectroscopy (EIS), from which the target device has a sheet resistance (R_{sh}) value of $185 \text{ k}\Omega$ compared to the $90 \text{ k}\Omega$ of the control device (Fig. S32, ESI†), indicating the inhibition of carrier recombination in the bulk and at the interface of the perovskite thin films. In addition, the high built-in electric field (V_{bi}) in the target device (1.11 V) over that in the control device (0.97 V) (Fig. S33, ESI†) suggests that less interface charge accumulation was found in the target device due to the reduced interface defect density. This was further confirmed by the J_{sc} versus light intensity and V_{oc} versus light intensity characteristics of the target device over the control device (Fig. S34 and S35, ESI†). The result indicates that the target device has less trap-assisted single-molecule recombination compared to the control device.

Conclusion

The δ -FAPbI₃ intermediate phase plays a crucial role in FAPbI₃ film preparation. However, the δ -FAPbI₃ intermediate phase is

usually characterized by chaotic crystal orientation and poor crystal quality, leading to a higher density of trapped states in the α -FAPbI₃ after phase transformation. We developed a tailoring processing temperature method to form high-quality δ -FAPbI₃ intermediate phase and thus obtaining the α -FAPbI₃ phase with lower density of trapped states. Through these improvements, we got perovskite solar cells reaching an efficiency of 25.05% with long operational stability over 1000 h. And it is worth noting that the trap state density of the α -FAPbI₃ thin film prepared by this method has decreased by about 8 times compared to traditional methods. However, in addition to the direct tailoring processing temperature method, more studies still need to be carried out to control the crystallization and phase transition processes of the δ -FAPbI₃ intermediate phase, to reduce the defects of the final α -FAPbI₃ perovskite and improve the performance of the PSC device.

Experimental procedures

Full details of experimental procedures can be found in the ESI.†

Conflicts of interest

There are no conflicts to declare.

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